Triplet Spectralwise Transformer Network for Hyperspectral Target Detection

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***Abstract*— Recently, deep learning methods have demonstrated their potentials in extracting spectral information for hyperspectral images and have been widely applied in hyperspectral target detection (HTD). However, prior deep learning methods, represented by the convolutional neural networks, mainly focus on the local information and representation, which cannot well capture the long-range dependence. Besides, limited target references cannot meet the need of massive labeled samples for the training process. This work develops a triplet spectralwise transformer-based target detector (TSTTD) to deal with these problems. First, this work explores a novel triplet spectralwise transformer network for the HTD task, and a data augmentation method is used to construct sufficient and balanced training samples for balanced learning. The proposed network shows advantages in learning local features from multiple adjacent bands and global features with long-range dependence. Second, for improving the separability between the targets and backgrounds, a novel intercategory separation and intracategory aggregation (ISIA) loss function is proposed, which joints the hard-negative-mining (HNM) triplet loss and the binary cross-entropy (BCE) loss. Third, the experimental results on six datasets show that our proposed method is effective in leading to excellent detection performance when compared with other state-of-the-art methods.**

***Index Terms*— Balanced learning, hyperspectral image, spectralwise transformer, target detection, triplet network.**

I. Introduction

HYPERSPECTRAL imaging is a landmark technology in the development of remote sensing. The generated images usually contain hundreds of narrow spectral bands over a wide range of the electromagnetic spectrum at each pixel [1], [2], [3]. The approximately continuous spectrum makes it possible to provide rich information for characterization of different materials spectrally. Therefore, hyperspectral images have been widely applied in many fields, such as land cover classification, military and civilian object detection, anomaly detection, and change detection [4], [5], [6], [7]. As an important hyperspectral analysis task, hyperspectral target detection (HTD) focuses on the detection and location of specific targets under limited target reference information [8], [17]. Such detection task still remains a challenging one due to the highly spectral variability of the observed spectra from the same material caused by inherent properties of the material, variations in environmental conditions, and sensor noise [9], [10], [11]. Besides, hybrid pixels formed by random mixing of target pixels and surrounding background pixels multiply the difficulties to discriminate the target pixel with the background [12]. Therefore, exploring discriminative methods to overcome these problems becomes urgent in target detection of the hyperspectral images.

The traditional hyperspectral target detectors mainly use the matching methods [13], such as spectral angular mapper (SAM) [14] and spectral information divergence (SID) [15], based on hand-crafted decision functions to recognize target objects from backgrounds. The good performance of the following classical methods is mainly guaranteed by the good prior assumptions [16], [17], such as distribution assumptions and energy minimization assumptions. The generalized likelihood ratio test (GLRT) [18] and its variants, such as adaptive coherence estimator (ACE) [19], [20] and adaptive matched filter (AMF) [21], are the representatives of distribution assumptions. In addition to these distribution assumptions, constrained energy minimization (CEM) [22] and its variants, such as target-constrained interference minimized filter (TCIMF) [23], construct the energy equations and discriminate the targets under energy minimization assumptions. Even though these methods are easy to implement, they usually cannot provide desired detection performance since these hand-crafted features are not discriminative enough to recognize the objects from the background. Besides, the complex practical application scenarios may not match the assumptions, placing restricts on the performance of classical methods.

Recently, deep learning methods have demonstrated their effectiveness in hyperspectral image classification [24], [25], [26], anomaly detection [27], [28], [29], and change detection [30], as well as in the literature of HTD tasks [35], [36], [37], [38], [39], [40], [42]. Generally, these methods possess multiple layers with large amounts of parameters and learn these parameters adaptively, and thus can capture subtle differences to discriminate between the targets and backgrounds. Among these deep learning methods for HTD, deep Siamese methods [31], [32], [33], [34], which have two identical networks that share the same weights, become an effective one since

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the Siamese architecture is fit for spectral matching. Some methods adopted Siamese fully connected neural network (FCNN) as the spectral embedding function and converted the similarities between the under-test spectra and the prior target spectrum to detection scores [35], [36]. In [37], a two-stream convolutional network target detector (TSCNTD) used the convolutional network (CNN) to extract spectral features, followed by a fully connected layer to transform subtracted features to detection results. In [38], a Siamese transformer target detector (STTD) used the transformer encoder to extract spectral features, and then the paired features were subtracted and fed into MLP for the final similarity scores. Besides, deep metric learning can also be applied in the training process of deep Siamese models to decrease the intraclass variance between different target spectra and increase the interclass variance between the target spectra and the background spectra.

Even though deep Siamese methods have shown their potentials in capturing discriminative features to discriminate between the targets and backgrounds, there exist several challenges when applied in the literature of HTD.

1) Large amounts of balanced sample pairs are required for the training process of deep Siamese models. Positive pairs and negative pairs can be constructed by pairing target samples and background samples with the prior target spectrum, respectively. While for the HTD task, limited target references cannot meet the need of massive target samples, and this would lead to the imbalanced and limited training samples.
2) Appropriate deep models should be designed to capture local features with discriminative characteristics and global features with long-range dependence for better recognition of the targets. Besides, a well-designed loss function is absolutely essential, contributing to improving the effect of separability between the targets and backgrounds so as to enhance the detection performance.

The key to solve the first problem lies in how to construct sufficient and balanced training samples using the single prior target spectrum and the hyperspectral image. Prior works to solve the problem are mainly divided into two categories: generation methods [39], [40] and synthesis methods [37], [38]. The first one generate training samples by generative models such as autoencoder [39] and generative adversarial network (GAN) [40]. Their generated target samples are relatively similar to the prior target spectrum that they fail to simulate the subtle spectral differences between target pixels. The latter one, synthesis methods, construct target samples by mixing the prior target spectrum and selected background samples [37], [38]. When it comes to the selection of background samples in [37], the random initialization of k-means causes large differences in clustering results, leading to fluctuations in performance. Besides, the imbalance of target–background classes in [38] is not conducive to the training process. Our work would focus on the synthesis methods and develops a data generation method based on the radiative transfer model [41]. Target samples are obtained by mixing the prior target spectrum with background samples. In this way, sufficient training samples are constructed with balanced target–background categories for balanced learning. We consider all the pixels of the HSI as background samples, ensuring the integrity and certainty of background samples, while improving the stability of the training process.

To solve the latter one, we propose a triplet spectralwise transformer network designed with a focus on hyperspectral characteristics to capture the long-range information. Most of the prior methods are mainly based on the CNNs to extract features of the hyperspectral image [37], [42]. These CNNs present their good performance in extracting spatially structural information and locally contextual information from the images [43], [44]. Nevertheless, the local receptive field mechanism in CNNs makes it difficult to learn long-range information of the spectral signal [45], [46]. To extract global spectral features, several works have used the transformers in the HTD task [47], [48]. Even though these prior transformers can capture the global information, the ignorance of locally spectral representations and the loss of important information in propagation would decrease the discrimination of the learned features. To further explore the potential of transformer in HTD, we propose a triplet spectralwise transformer network which can solve the prior problems with the group-based spectral embedding (GSE) [49] and skip-layer concatenation fusion (SCF) operation [49]. Besides, our triplet spectralwise transformer can make full use of the balanced training samples and thus extract more discriminative features from the image.

In addition to the triplet spectralwise transformer, we propose a novel intercategory separation and intracategory aggregation (ISIA) loss which joints the hard-negative-mining (HNM) triplet loss and the binary cross-entropy (BCE) loss. In the aforementioned detectors, the loss function can be divided into two classes, including classification loss [35], [38] and contrastive loss [37]. Classification loss considers target detection as a binary classification problem, such as BCE loss [35] and focal loss [38], which only uses the pixelwise label for training but fails to take full use of the extracted feature pairs. Contrastive loss focuses on the penalization of pairwise correlation of extracted features [34], [52], [53], [54]. However, the contrastive loss in [37] only considers the correlation between samples and the prior target spectrum while ignoring the correlation between target samples and background samples. Furthermore, considering the pairwise correlation makes the training process vulnerable to imbalanced and limited training samples. This would increase the randomness and uncertainty of the training process. To solve these problems, this work proposes the ISIA loss with the HNM triplet loss [57] to aggregate positives and anchor as well as separate negatives from anchor and positives, and the BCE loss to accelerate the learning convergence of the deep model.

Considering the merits of both the data augmentation and deep-learning-based methods, this work proposes a triplet spectralwise transformer-based target detector for hyperspectral images, denoted as TSTTD. First, we constructed sufficient and balanced training samples based on the radiative transfer model. Then, the triplet spectralwise transformer network is developed with shared parameters. Furthermore, we propose an ISIA loss which joints the HNM triplet loss and the

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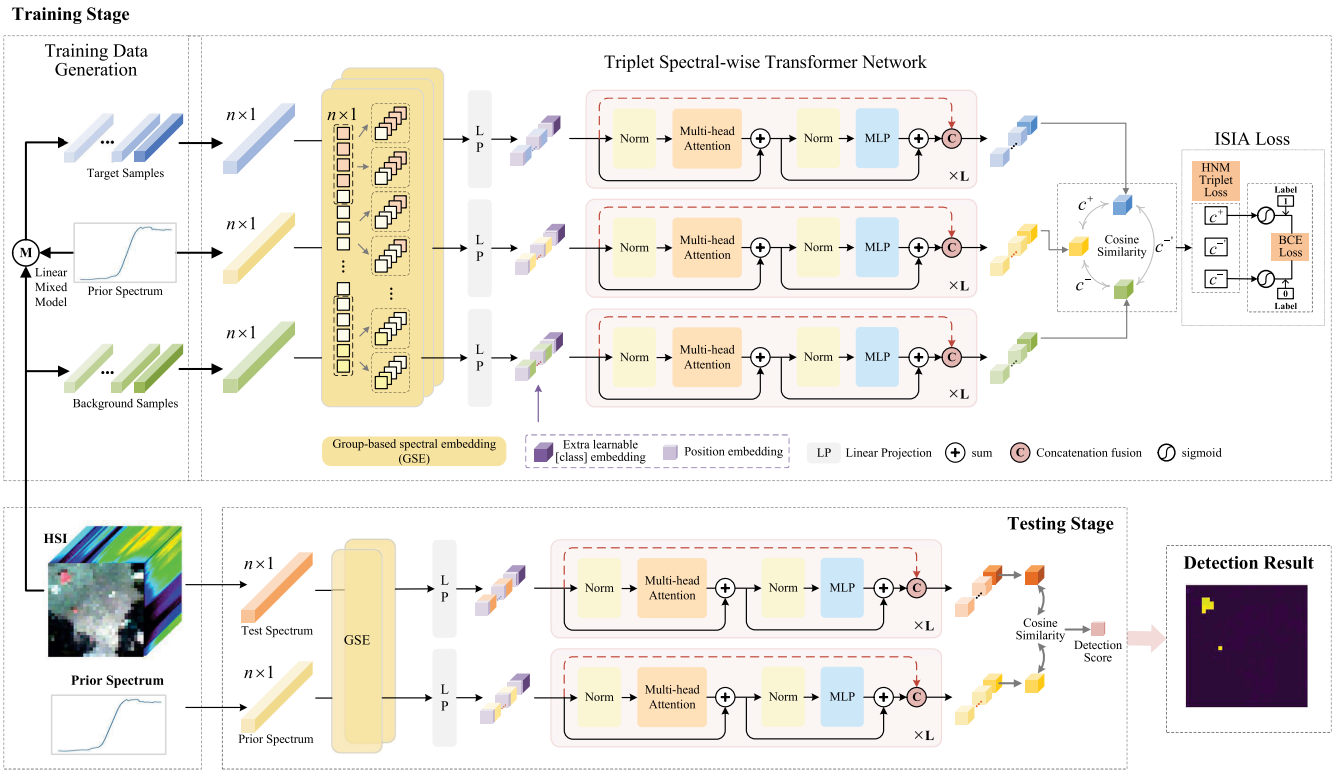


Fig. 1. Overview illustration of the proposed TSTTD for HTD.

BCE loss to enlarge the separability between the targets and backgrounds for better detection performance. The main contributions of the proposed method are summarized as follows.

1) This work explores a TSTTD for HTD. The triplet spectralwise transformer network can highlight the targets and suppress backgrounds with the spectral similarity metric learning. Besides, the data augmentation method can provide sufficient and balanced training samples which can further improve the detection performance.
2) This work proposes an ISIA loss for the HTD task, which can take advantage of the correlation between different training samples and the pixelwise label. The ISIA loss contributes to enlarging separability of targets and backgrounds, leading to excellent detection performance.
3) This work conducts comparison experiments and ablation experiments on six datasets to verify the effectiveness of the developed method.

II. Proposed Method

Given a specific hyperspectral image, HTD tries to detect and locate the specific target under a given prior target spectrum. In this work, a TSTTD is proposed. The following will introduce the proposed method in detail.

A. Overall Framework

The framework of the TSTTD is shown in Fig. 1. The training process consists of three parts: training data generation, triplet transformer, and ISIA loss. First, sufficient and balanced training samples are generated using the prior target spectrum and the hyperspectral image. Then the target samples, the prior target spectrum, and the background samples are fed into the triplet spectralwise transformer network. The proposed network integrates the GSE and the SCF. The network is optimized using the proposed ISIA loss. The details of the proposed TSTTD are described as follows.

B. Construction of Training Samples

In the HTD method based on deep learning, there exists a severe problem of inadequate training samples. This work develops a data generation method to construct sufficient and balanced training samples including background training samples and target training samples, using only the hyperspectral image itself and the limited prior information. The proposed data generation method is inspired by the radiative transfer model [41]. Hyperspectral sensors sample the electromagnetic radiation of the specific region across the visible and near infrared. A radiative transfer model expresses the radiance transfer between the observed ground materials and the sensor. A frequently used model for the sensor-measured spectral radiance of the observed pixel $L_s(\lambda)$ is

$$L_s(\lambda) = L_{\mathrm{TGR}}(\lambda) r(\lambda) + L_{\mathrm{adj}}(\lambda) r_a(\lambda) + L_u(\lambda) \quad (1)$$

where all the terms depend on the spectral band represented by wavelength λ and are defined as follows.

1) $L_{\mathrm{TGR}}(\lambda)$ is the total ground-reflected (TGR) spectral radiance. It is for all the sources of radiation with different paths in the atmosphere that are reflected at the observed target.

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2) $r(\lambda)$ is the reflectance of the observed target.
3) $L_{\text{adj}}(\lambda)$ is part of the radiation, which is caused by the adjacency effect.
4) $r_a(\lambda)$ is the average spectral reflectance of the surrounding region of the target.
5) $L_u(\lambda)$ is the spectral path radiance, which is scattered in the atmosphere directly into the sensor without being reflected by the ground.

In conclusion, the sensor-measured radiance of the observed pixel originates from three main parts: the target reflection, the adjacent materials' reflection, and the atmospheric scattering.

Considering target reflection and adjacency effect, we construct target training samples L_s^{target} by mixing the prior target spectrum L_s^{prior} and the background spectrum L_s^b in random proportions

$$L_s^{\text{target}} = (1-\alpha) \times L_s^{\text{prior}} + \alpha \times L_s^b. \tag{2}$$

α is the abundance of background, which is randomly set as $0\% - -10\%$ in a pixel. All the spectra in the under-test HSI with N pixels are considered as background samples L_s^b to estimate the background information of the image. The path radiance is concluded in the in-scene background spectru L_s^b. The reference spectrum of the target L_s^{prior} can be obtained from a spectral library or from target pixels in the test HSI. In this way, subpixels in real scenes can be formulated by mixing target pixels with the surrounding background pixels. For a given prior target spectrum, N target samples can be generated by N background samples. Therefore, sufficient and balanced background samples and target samples can be obtained.

Then we can construct two types of training sample pairs using these obtained samples and the prior target spectrum to optimize the Siamese network framework. One type is the positive training sample pairs constructed by the prior target spectrum and target samples, and the other type is the negative training samples constructed by the prior target spectrum and background samples. For the given prior target spectrum, the positive training sample pairs and the negative training sample pairs have the same number, which indicates a balanced distribution in the training stage.

In the training stage, the label of the positive sample pair combined by the prior target spectrum and a target sample is $y_i^+ = 1$, and the label of the negative sample pair combined by the prior target spectrum and a background sample is $y_i^- = 0$. In the testing stage, each test spectrum in the HSI with N pixels is paired with the prior target spectrum. N testing pairs are fed into the well-trained network to get the detection scores of N under-test spectra.

C. Triplet Spectralwise Transformer

The proposed triplet spectralwise transformer network transforms the target detection task into the spectral matching task, solved by learning a similarity metric. In our case, learning the similarity metric is realized by constructing a triplet architecture consisting of three identical networks that share the same parameters. Moreover, the proposed network is designed with a focus on hyperspectral characteristics based on the transformer. It integrates the GSE [49] to extract locally spectral characteristics from multiple adjacent bands and the SCF [49] to concatenate and fuse residuals, reducing information loss from shallow to deep layers.

1) Group-Based Spectral Embedding: Hyperspectral sensors measure the radiation of the specific region at a detailed wavelength range, giving the spectrum composed of hundreds of narrow contiguous spectral bands. Each hyperspectral pixel corresponds to a spectrum, with different values for different bands in the spectral dimension. This detailed spectral information can be used to analyze different objects. Capturing the local band-to-band variability of such spectrum helps effectively discriminate target pixels from the backgrounds in the under-test hyperspectral image. Similar to the typical transformer, we use the learnable embedding module to convert input tokens into embeddings.

Specifically, we take groups of several adjacent bands of the spectrum as input tokens to capture and characterize local spectral properties. We use an overlapping sliding window represented by the function to decompose the spectrum into the grouped representation $g(\cdot)$ to decompose the spectrum $s = [s_1, s_2, \ldots, s_n]^{\text{T}} \in \mathbb{R}^{n\times 1}$ into the grouped representation

$$\boldsymbol{X} = g(s) = [x_1, \ldots, x_i, \ldots, x_n]^{\text{T}} \in \mathbb{R}^{n\times m} \tag{3}$$

where $x_i = [s_{i-\lfloor m/2 \rfloor}, \ldots, s_i, \ldots, s_{i+\lfloor m/2 \rfloor}]^{\text{T}} \in \mathbb{R}^{m\times 1} (i = 1, 2, \ldots, n)$, and $\lfloor \cdot \rfloor$ indicates the rounding down operation. $\boldsymbol{X}$ is composed of n token vectors of length m. The scalars n and m represent the number of bands of the spectrum and the number of considered adjacent bands, respectively. The scalar n also serves as the input sequence length for the transformer encoder.

Then the linear projection is used to convert the grouped representation $\boldsymbol{X} = \mathbb{R}^{n\times m}$ into feature embeddings $\boldsymbol{Z} = \mathbb{R}^{n\times d}$ of d dimensions. As class token in Vision Transformer [48], we prepend learnable class embedding to the feature embeddings of spectral groups, whose output in the transformer encoder works as the spectral representation. Besides, to preserve position information, position embeddings are added to the feature embeddings. Finally, the resulting embeddings $\boldsymbol{Z} = \mathbb{R}^{(n+1)\times d}$ with feature embeddings, class embedding, and position embeddings act as the input to the transformer encoder.

2) Skip-Layer Concatenation Fusion: The proposed triplet network consists of three transformer encoders with shared structures and parameters. The transformer encoder is composed of several identical layers. Each layer consists of a multihead self-attention (MSA) block and a fully connected feedforward network (FFN) block. Layernorm is used before each block, and residual connection around each of the two blocks. The FFN block contains two layers with a Gaussian error linear unit (GELU) activation.

Skip connection is a widely used strategy to improve the propagation of information from shallow to deep layers of deep neural networks and reduce information loss in the learning stage. In particular, shortcut connections mentioned in [55] are those that skip one or more layers. In our case, based on the typical transformer encoder layers, we design shortcut connections simply skipping one layer to improve

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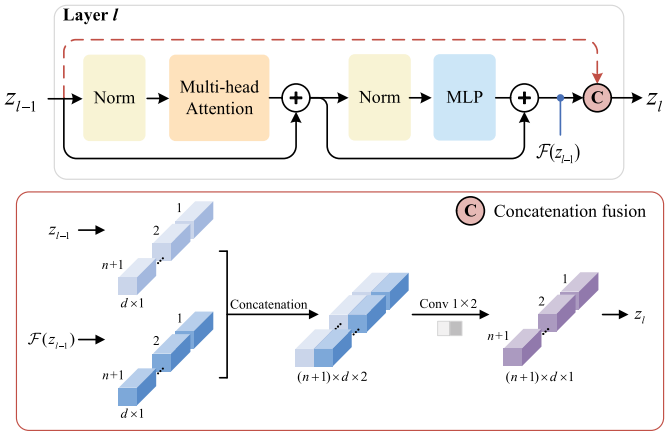


Fig. 2. Illustration of SCF in the proposed TSTTD.

the propagation of information between layers by learning to concatenate and fuse the residuals.

The transformer encoder typically consists of L identical layers, each layer implementing a nonlinear mapping, where l indexes the layer. We denote the input and output of the lth layer as $z_{l-1} \in \mathbb{R}^{(n+1)\times d}$ and $z_l \in \mathbb{R}^{(n+1)\times d}$, respectively. Specifically, z_{l-1} is the input of the whole encoder if $l = 1$. The shortcut connection is defined as follows, as z_{l-1} and the output of $\mathcal{F}_l(\cdot)$ are combined by concatenation and fusion

$$z_l = W_l\left[\mathcal{F}_l(z_{l-1}), z_{l-1}\right] \tag{4}$$

where $[\mathcal{F}_l(z_{l-1}), z_{l-1}]$ refers to the concatenation operation of features, and W_l denotes learnable parameters of a 1×2 convolutional layer for fusion operation. Finally, the output vector corresponding to the class embedding of the last transformer encoder layer serves as the spectral feature representation. The illustration of SCF in the proposed TSTTD is presented in Fig. 2.

D. ISIA Loss Function

Learning from triplets means training with samples of the form. $\boldsymbol{p}_i$ is the generated target sample called positive sample, $\boldsymbol{t}_i$ is the prior target spectrum called anchor, and $\boldsymbol{n}_i$ is the selected background sample called negative sample. Three samples in each triple $\{\boldsymbol{p}_i, \boldsymbol{t}_i, \boldsymbol{n}_i\}$ specifically pass through three branches of the triplet transformer network at the same time, giving vectors $\{f(\boldsymbol{p}_i), f(\boldsymbol{t}_i), f(\boldsymbol{n}_i)\}$ to represent the extracted spectral feature representations. Typically, optimizing the parameters of a triplet network brings $f(\boldsymbol{p}_i)$ and $f(\boldsymbol{t}_i)$ close and pushes $f(\boldsymbol{n}_i)$ and $f(\boldsymbol{t}_i)$ far apart.

Generally, the triplet network usually uses the margin ranking loss function [56]. This loss drives that the anchor is closer to the positive samples than it is to the negative samples, while the difference between the positive difference and negative difference requires to be at least an enforced margin value. However, there are some negative samples that are far away from the anchor, making the loss easily converge to 0, and thus, the weights can stop updating easily, which makes a limited contribution for training. Since the margin ranking loss ignores the negative distance between the negative and positive samples, the in-triplet hard negative mining method proposed in [57] is taken into account. The negative distance between the negative and positive samples can therefore contribute to improving the model, so that the learning process can not only push the negative samples farther from the anchor in the feature space but also the positive samples.

Considering the characteristics of HTD task and the instability of training process caused by different scales, this work develops a scale-invariant HNM triplet loss for HTD. To implement scale-invariant, this work uses the cosine similarity as measurement to formulate the loss. The cosine similarity is described as follows:

$$c(u, v) = \frac{u \cdot v}{\|u\|_2 \times \|v\|_2} \tag{5}$$

where u and v are two vectors of the same dim, and $u \cdot v$ denotes the dot product. The absolute value of $c \in [-1, 1]$, $|c|$, is regarded as the similarity score of two vectors; the closer it is to 1, the closer the two vectors, otherwise further apart. Based on the cosine similarity mentioned above, the vectors in each triplet $\{f(\boldsymbol{p}_i), f(\boldsymbol{t}_i), f(\boldsymbol{n}_i)\}$ are combined in pairs to obtain three possible similarity scores including one positive c_i^+ and two negatives c_i^- and $c_i^{-'}$, where

$$\begin{cases} c_i^+ = \left|c\left(f\left(\boldsymbol{p}_i\right), f\left(\boldsymbol{t}_i\right)\right)\right| \\ c_i^- = \left|c\left(f\left(\boldsymbol{n}_i\right), f\left(\boldsymbol{t}_i\right)\right)\right| \\ c_i^{-'} = \left|c\left(f\left(\boldsymbol{n}_i\right), f\left(\boldsymbol{p}_i\right)\right)\right|. \end{cases} \tag{6}$$

The largest negative similarity should be smaller than the positive similarity by a margin value. Therefore, we define the largest negative similarity within the triplet as the in-triplet hardest negative, denoted by $c_i^* = \max(c_i^-, c_i^{-'})$. This ensures that the hardest negative inside the triplet is used for backpropagation. Subsequently, the HNM loss function becomes

$$L_{\text{triplet}} = \frac{1}{b}\sum_{i=1}^{b}\max\left(0, \mu + c_i^* - c_i^+\right) \tag{7}$$

where b represents the batch size. In this way, the hardest negative within the triplet is used for backpropagation. The goal of the loss function is to maximize c_i^+ while minimizing c_i^- and $c_i^{-'}$, leading to $c_i^+ < \mu + c_i^*$. Minimizing the loss function drives the positive sample close to the prior target in the feature space and pushes the negative sample away from both the prior target and the positive sample. It is worth noting that forcing a minimal separation μ between the negative similarity and the positive similarity helps the discrimination between positive and negative samples.

However, the HNM triplet loss takes advantage of the correlation between different samples while ignores the pixelwise label. Therefore, BCE loss is introduced to supervise the training process. The similarity label of the positive sample and the negative sample is 1 and 0, respectively. Given the similarity scores c_i^+ with label $y_i^+ = 1$ and c_i^- with label $y_i^- = 0$, the BCE loss function is formulated as follows:

$$\begin{aligned} L_{\text{BCE}} = -\frac{1}{2b}\sum_{i=1}^{b} & y_i^+ \cdot \log\left[\sigma\left(c_i^+\right)\right] \\ & + \left(1 - y_i^-\right) \cdot \log\left[\sigma\left(1 - c_i^-\right)\right] \end{aligned} \tag{8}$$

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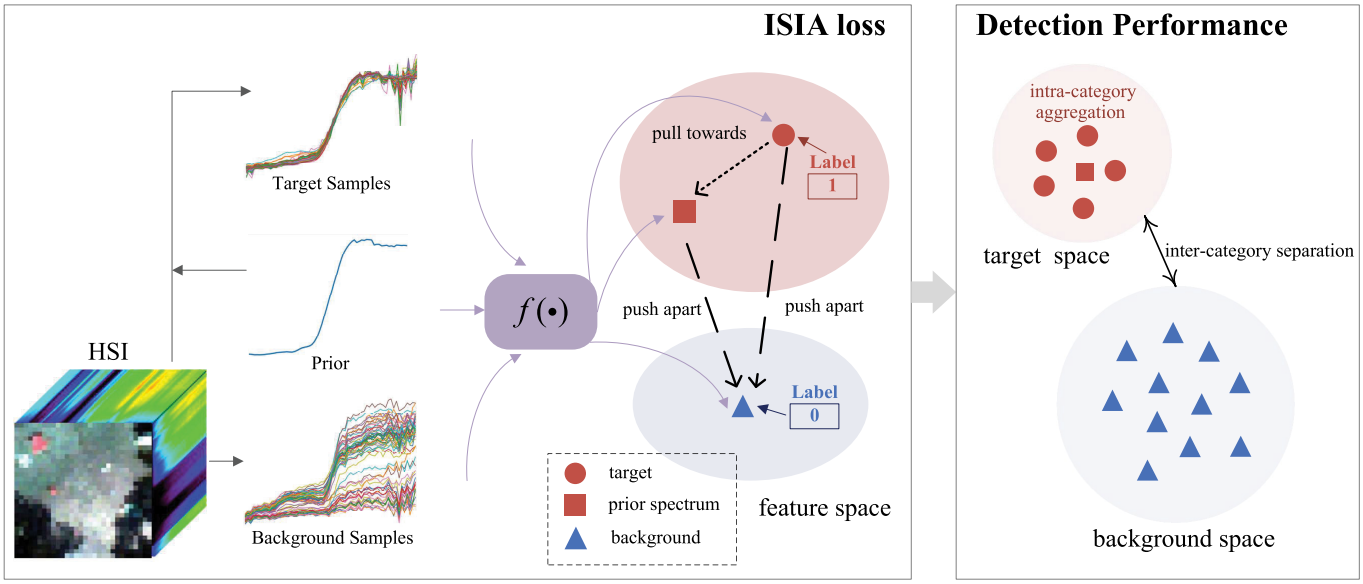


Fig. 3. Illustration of the proposed ISIA loss, which joints the HNM triplet loss and BCE loss. The ISIA loss enables aggregation of positives, which pulls target samples toward the prior spectrum. Besides, the ISIA loss enables separation of negatives, which not only pushes background samples and the prior target spectrum apart but also pushes them apart from target samples. Moreover, the pixelwise label is used to supervise the training process. The ISIA loss can contribute to enlarging the separability of targets and backgrounds, improving the detection performance.

where $\sigma(\cdot)$ is the sigmoid function, and b is the batch size. Each mini-batch includes b negatives samples and b corresponding positive samples. L_{BCE} is used to measure the distance between the similarity scores and the labels.

To be summarized, this work develops a novel ISIA loss which joints HNM triplet loss and BCE loss. The ISIA loss function L is the sum of L_{Trilpet} and L_{BCE}

$$L = L_{\mathrm{Triplet}} + L_{\mathrm{BCE}}. \tag{9}$$

The illustration of the ISIA loss is presented in Fig. 3.

III. Experimental Results and Analysis

A. Experimental Datasets

In our experiments, six images from four public hyperspectral datasets are used to validate the proposed method.

1) MUUFL Gulfport Dataset: In the MUUFL Gulfport dataset, the hyperspectral data were acquired using a Compact Airborne Spectrographic Imager (CASI-1500) over the Gulf Park Campus of the University of Southern Mississippi. It has 325 × 337 pixels with 72 spectral bands, and the scene and scene labels are publicly available in [58] and [59]. After removing the noisy bands (bands no. 1–4 and 69–72) [59], a subimage of 30 × 30 pixels and 64 bands is selected from the original HSI data [60]. In this subimage, two solid brown cloth panels are selected as targets, and the target spectrum with laboratory spectrometer measurements is known.

2) Airport-Beach-Urban Dataset: The second dataset is the airport-beach-urban (ABU) dataset [6]. There are four airport scene images, four beach scene images, and five urban scene images in the ABU dataset. We select one image from the beach scene and two images from the urban scene, and each of these selected images was captured by airborne visible/infrared imaging spectrometer (AVIRIS) with a spatial size of 100 × 100 pixels. In the ABU dataset [6], we select one beach scene image and two urban scene images as representative datasets for experiments, named Beach, Urban1, and Urban2, respectively. Beach has 193 bands, and 202 pixels are regarded as targets. Urban1 has 204 bands, containing 67 pixels of nine buildings as targets. Urban2 has 207 bands, and 88 pixels of nine buildings are regarded as targets.

3) San Diego Dataset: The third dataset was captured by AVIRIS over the San Diego airport area, CA, USA. The San Diego dataset contains 100 × 100 pixels, and each pixel has 189 bands after removing 35 water vapor absorption and low SNR bands. Precisely, bands no. 1–6, 33–35, 97, 107–113, 153–166, and 221–224 are discarded [61]. Three aircrafts consisting of 134 pixels are regarded as targets to be detected in the scene.

4) HYDICE Dataset: The fourth dataset was captured by the hyperspectral digital imagery collection experiment (HYDICE) sensor over an urban area, CA, USA. The size of the scene is 80 × 100 pixels, and each pixel has 175 bands [6].

B. Assessment Criteria

One qualitative and two quantitative criteria are used to evaluate the performance of our proposed target detection method.

1) Receiver Operating Characteristic Curve: The receiver operating characteristic curve (ROC) is used to qualitatively evaluate the performance of the detector. The ROC curve is a plot of the true positive rate (TPR) versus the false positive rate (FPR) for different threshold settings on the detection probability map. The vertical coordinate of the ROC curve is TPR, also known as the probability of detection. The

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TABLE I
AUC VALUES' COMPARISON FOR SIX DATASETS

Methods	Muufl Gulfport	Urban1	Urban2	Beach	San Diego	HYDICE
SAM	0.95483	0.99478	0.99904	0.92762	0.99677	0.95783
SID	0.71449	0.99441	0.93118	0.98373	0.99684	0.94662
ACE	0.62483	0.92878	0.99218	0.86418	0.96848	0.90976
CEM	0.79371	0.94035	0.99791	0.84853	0.98689	0.94694
MF	0.82685	0.93566	0.99739	0.84691	0.98503	0.95239
SFCTD	0.94169	0.99178	0.99548	0.95215	0.99055	0.99648
TSCNTD	0.93427	0.98127	0.99535	0.76545	0.77366	0.88615
Ours	**0.99843**	**0.99787**	**0.99958**	**0.99507**	**0.99787**	**0.99966**

horizontal coordinate of the ROC curve is FPR, also known as the probability of false alarm. TPR and FPR are defined as follows:

$$\mathrm{TPR} = \frac{N_c}{N_t} \quad \mathrm{FPR} = \frac{N_f}{N_b} \tag{10}$$

where N_c is the number of correctly detected target pixels (actually target, predicted to be target), N_t is the total number of real target pixels, N_f is the number of false alarm pixels (actually background, predicted to be target), and N_b is the total number of background pixels [62]. A well-performing detector with high TPR and low FPR corresponds to an ROC curve close to the top left.

2) Area Under the ROC Curve: The area under ROC curve (AUC) can quantitatively measure the performance of the detector, which is convenient for comparing and evaluating different detectors. The AUC value can be obtained by integrating the ROC curve along abscissa, and it varies between 0 and 1. The AUC value can be defined as follows:

$$\mathrm{AUC} = \int_{-\infty}^{+\infty} \mathrm{TPR}(t)d\,\mathrm{FPR}(t) = \int_{-\infty}^{+\infty} \mathrm{TPR}(t)\,\mathrm{FPR}'(t)dt \tag{11}$$

where t represents the threshold for the detection probability map [63]. A well-performing detector corresponds to an ROC curve close to the top left corner, indicating a relatively large AUC value.

3) Separability Map: The separability map in the form of boxplots can effectively assess the detector's separation of the targets against the backgrounds and the suppression of the backgrounds. The separability map consists of a red box with whiskers and a blue box with whiskers, which represents the distribution of detection values of target pixels and background pixels, respectively. The upper and lower quartiles of the box are the first and third quartiles of the data, respectively. The upper and lower edges of box whiskers represent the maximum and minimum of the data, respectively. In HTD, the distance between the red box and the blue box represents how well the detector separates the targets from the backgrounds; the greater the distance, the better the separation. Moreover, the lower the blue box, the better the detector's suppression of the backgrounds.

C. Experimental Setups

To validate the effectiveness of the proposed TSTTD, five widely used classical methods and two advanced deep-learning-based methods are compared with TSTTD. Five classical methods [17] include SAM, SID, ACE, CEM, and MF. Two deep-learning-based methods include FCNN-based SFCTD [35] and CNN-based TSCNTD [37]. The settings of deep-learning-based methods for comparison are consistent with the original work. For all the methods, the prior spectrum in each dataset is the same. The target prior spectrum for Muufl Gulfport was obtained from laboratory spectrometer measurements, and the prior target spectrum of other datasets was obtained from target pixels in the test HSI.

For the proposed triplet transformer network, we use the Adam optimizer for all the experiments, the batch size is 64, the learning rate is 0.0001, and the weight decay is 0. For construction of training samples, the abundance of background in the linear mixed model is randomly set as 0%–10% in a pixel. All the experiments are run on PyTorch 1.9.0 with a CPU of Intel Core i9-10900X at 3.70 GHz, 32-GB RAM, and a GPU of NVIDIA GeForce RTX 3090.

D. Comparison and Analysis of Experimental Results

In this section, we validate the effectiveness of the proposed TSTTD and compare its performance with different methods. Table I shows the AUC values of different images to quantitatively measure the detection performance. Figs. 4–15 show the detection maps, ROC curves, and separability maps on different comparison methods on six datasets.

For Muufl Gulfport, the false color image and ground-truth map are shown in Fig. 4(a) and (b), and the color maps of detection results for different detectors are shown in Fig. 4(c)–(j). The first five detectors are terrible at separating targets from backgrounds or suppressing the background pixels. An important reason for this result is that the prior target spectrum was measured in the laboratory and is dramatically different from those in real scenarios. These classical methods completely ignore the target in the lower part of the screen, and they highlight only small portions of the top left target. Although SFCTD and TSCNTD achieve good performance at highlighting pixels of interest, the separation of the target pixels from their surrounding background pixels is inferior, exhibiting ambiguous boundaries of the target. A large number of background pixels are detected with high confidence by SFCTD and TSCNTD, decreasing the overall detection performance. It is evidently clear that our proposed detector, TSTTD, achieves excellent detection results, exhibiting extremely high level of visual contrast. TSTTD can not only highlight the target pixels but also strongly suppress the background pixels. The corresponding separability map and

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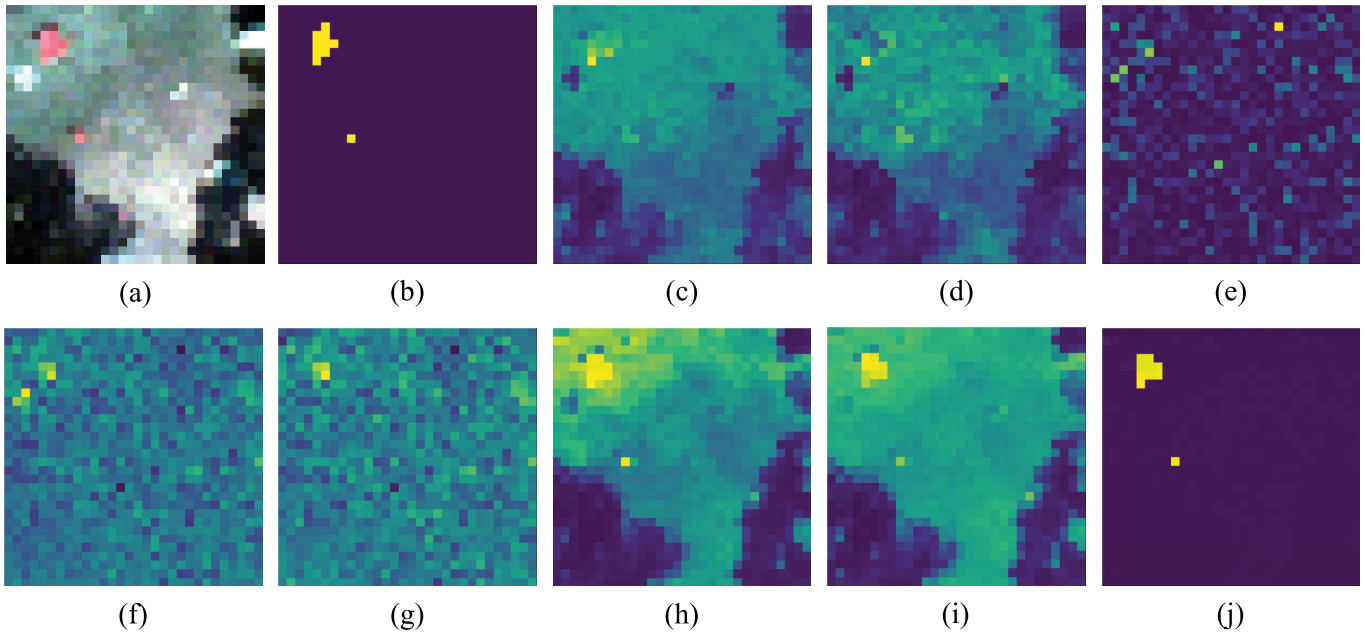
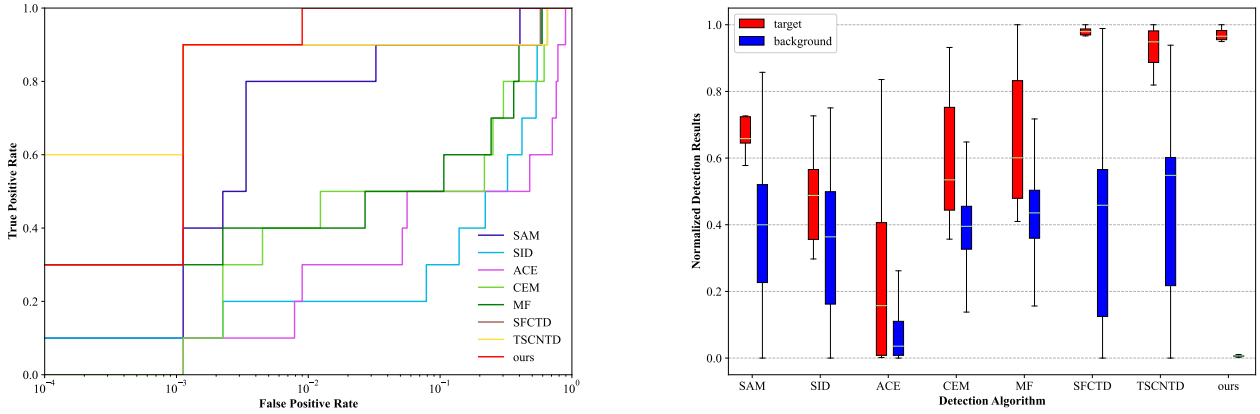


Fig. 4. Muufl Gulfport scene and detection maps of different comparison methods. (a) False color image. (b) Ground truth. (c) SAM. (d) SID. (e) ACE. (f) CEM. (g) MF. (h) SFCTD. (i) TSCNTD. (j) Ours.



(a) (b)

Fig. 5. ROC curves and separability map on Muufl Gulfport. (a) ROC curves. (b) Separability map.

ROC curves in Fig. 5 prove the above conclusions. In the separability map, it is clear that TSTTD has the lowest and flattest blue box that is extremely close to 0, which means the outstanding performance in suppressing backgrounds. TSTTD also has an extremely high red box, which means the excellent performance in highlighting targets. The blue and red boxes of TSTTD are completely separate with an extremely large gap, indicating remarkable separation of targets from backgrounds. The ROC curves of TSCNTD and our proposed TSTTD are closer to the top-left corner, indicating good detection performance. When all the targets are truly detected by TSTTD, the proportion of the backgrounds that are mistaken for targets is below 10^{-2}, representing the most competitive false alarm rate. Then the AUC values are provided in Table I to quantitatively assess the detection performance. Our proposed TSTTD achieves the best AUC value 0.99843, which is 0.0436 higher than the second-best SID of 0.95483.

For Urban1, the false color image and ground-truth map are shown in Fig. 6(a) and (b), and the maps of detection results for different detectors are shown in Fig. 6(c)–(j). All the five classical detectors fail to highlight the two buildings located on the upper boundary. SID and ACE show good background suppression capability, but they detect targets with low confidence. SFCTD and TSCNTD can better highlight all the buildings and maintain the morphological integrity of targets to some extent, but they falsely detect backgrounds with high confidence, causing decreased detection performance. When it comes to our proposed detector, TSTTD, it maintains the morphological integrity of all the targets with excellent confidence, while being able to suppress the detection scores of backgrounds to an extremely low level, exhibiting remarkably high visual contrast. The corresponding separability map and ROC curves are presented in Fig. 7. It is clear that in the separability map, the red box of TSTTD has the highest

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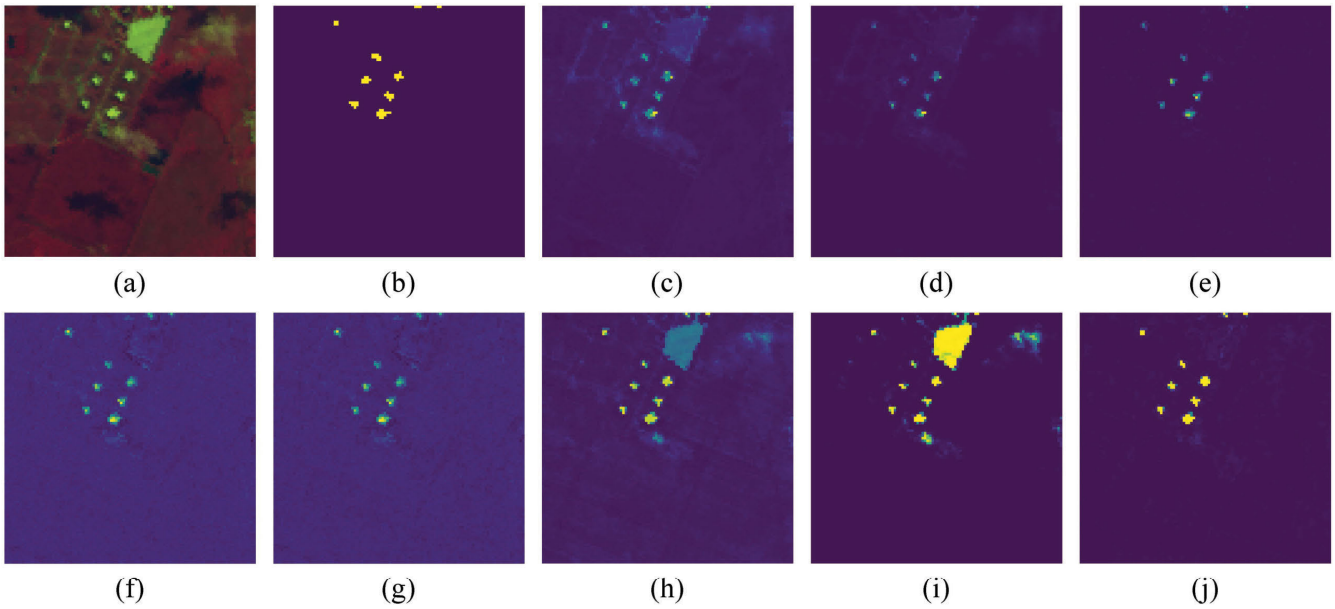


Fig. 6. Urban1 scene and detection maps of different comparison methods. (a) False color image. (b) Ground truth. (c) SAM. (d) SID. (e) ACE. (f) CEM. (g) MF. (h) SFCTD. (i) TSCNTD. (j) Ours.

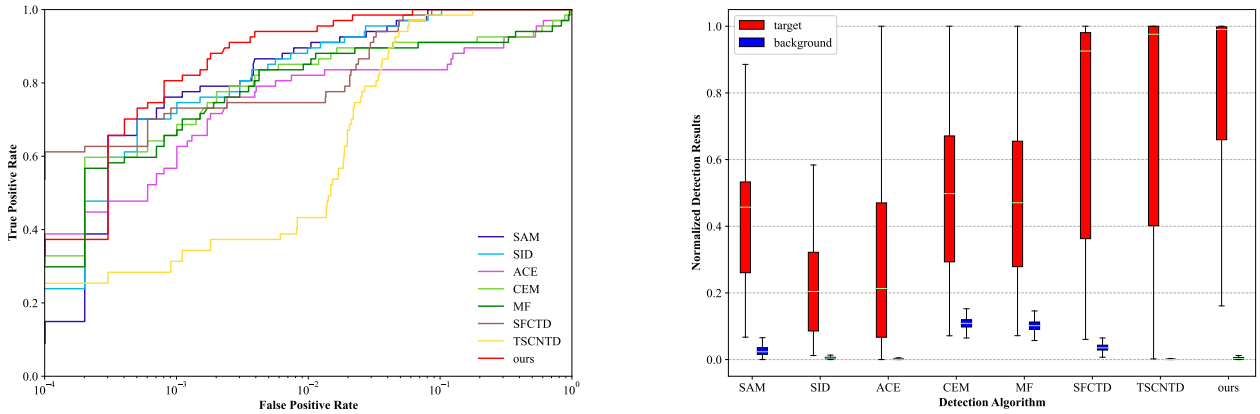


Fig. 7. ROC curves and separability map on Urban1. (a) ROC curves. (b) Separability map.

quartile and median, which means the excellent performance in highlighting targets. The blue and red boxes of TSTTD are completely separate with no overlaps, indicating separation of targets from backgrounds. The ROC curves of detectors except TSCNTD are close to the top-left corner, representing good detection performance. The red curve is the ROC curve of our proposed TSTTD, which shows the competitive detection rate under many FPRs. Then the AUC values are provided in Table I to quantitatively assess the detection performance. Our proposed TSTTD achieves the best AUC value 0.99787 and outperforms the second-best SAM of 0.99478 by 0.00309.

For Urban2, the false color image and ground-truth map are shown in Fig. 8(a) and (b), and the maps of detection results for different detectors are shown in Fig. 8(c)–(j). The classical detector, SID, is poor at separating targets from backgrounds, exhibiting low visual contrast. SAM performs better in locating targets and suppressing backgrounds and can also maintain the shape of targets, but the targets are not obvious. ACE achieves remarkable performance in suppressing backgrounds, but it only highlights the middle pixels of targets. CEM and MF perform better in highlighting most of the targets and show good overall separation to some extent, but they fail to suppress backgrounds. SFCTD and TSCNTD achieve extremely high detection scores of targets and exhibit high visual contrast, but too much emphasis is placed on highlighting targets that many background pixels are detected with extremely high scores by mistake. Worthy of special mention is that our proposed detector, TSTTD, highlights and maintains the morphological integrity of targets, and it is effective in suppressing backgrounds, showing high visual contrast. The corresponding separability map and ROC curves are presented in Fig. 9. In the separability map, it is obvious that the red boxes for TSCNTD and TSTTD are exceedingly high and flat, which means that they perform superbly in highlighting targets. The blue and red boxes of TSTTD are completely separate with a larger gap than other detectors, which indicates

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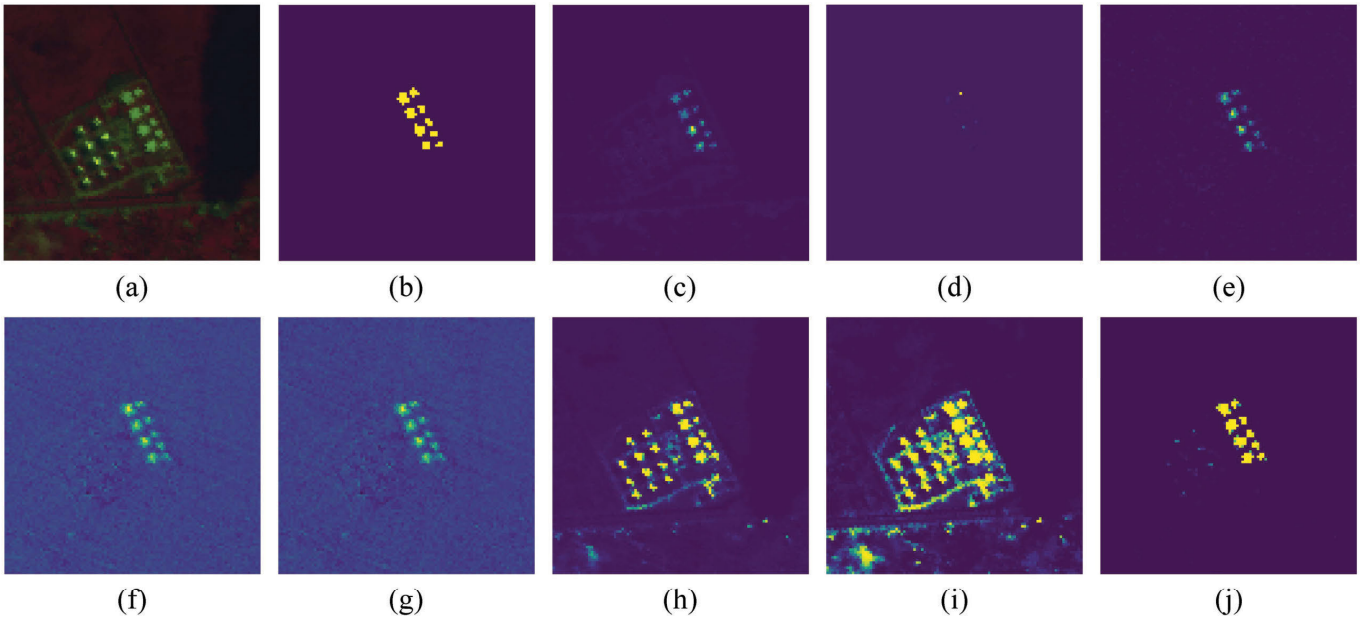


Fig. 8. Urban2 scene and detection maps of different comparison methods. (a) False color image. (b) Ground truth. (c) SAM. (d) SID. (e) ACE. (f) CEM. (g) MF. (h) SFCTD. (i) TSCNTD. (j) Ours.

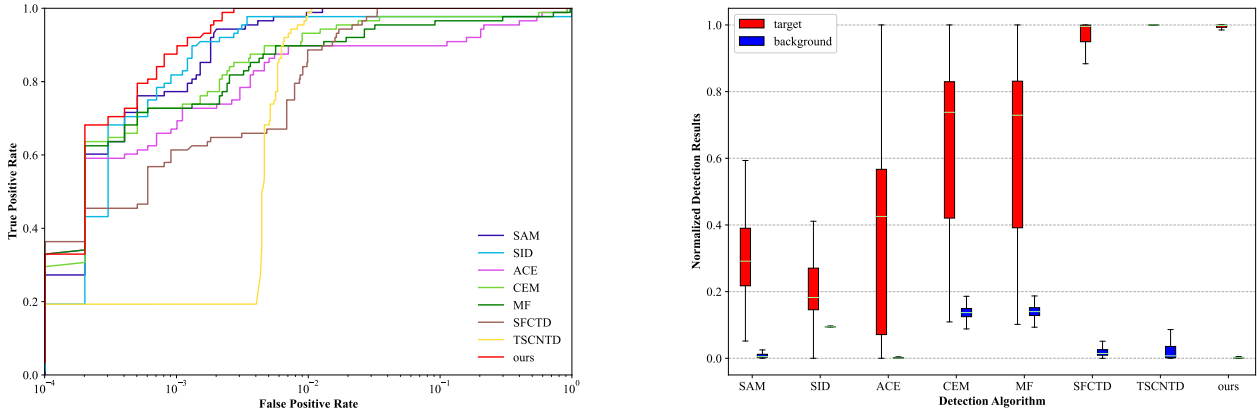


Fig. 9. ROC curves and separability map on Urban2. (a) ROC curves. (b) Separability map.

the best separation of targets from backgrounds. The ROC curve of our proposed TSTTD is the closest to the top-left corner, representing high target detection rates and low false alarm rates under different thresholds. Moreover, when all the targets are truly detected, the proportion of backgrounds that are mistaken for targets is far below 10^{-2}, representing the lowest false alarm rate, indicating excellent background suppression. Then the AUC values are provided in Table I to quantitatively assess the detection performance. Our proposed TSTTD achieves the best AUC value 0.99958 and the second-best SAM achieves 0.99904. The learning-based SFCTD and TSCNTD achieves lower AUC values than the classical SAM, CEM, and MF.

For Beach, the false color image and ground-truth map are shown in Fig. 10(a) and (b), and the maps of detection results for different detectors are shown in Fig. 10(c)–(j). We find that SAM, SID, ACE, and TSCNTD fail to highlight targets, showing extremely low visual contrast. The effects of CEM and MF on highlighting the target pixels are significantly better, but they are not very effective at suppressing the background pixels. SFCTD tends to achieve high detection scores at target pixels, but it wrongly detects the strong edge area with high confidence, decreasing the detection performance. In contrast, our proposed method, TSTTD, is less attracted by backgrounds with strong edge that it almost does not detect backgrounds as targets by mistake. The corresponding separability map and ROC curves are presented in Fig. 11. The separability map can prove the above conclusions. CEM and MF have the highest and widest blue boxes, and the median lines of the red boxes are far below 0.6, indicating poor background suppression and target detection. The red box for our proposed TSTTD has the highest upper quartile, median, and lower quartile, which means the best performance in target detection. The lower quartile of the red box for TSTTD is far

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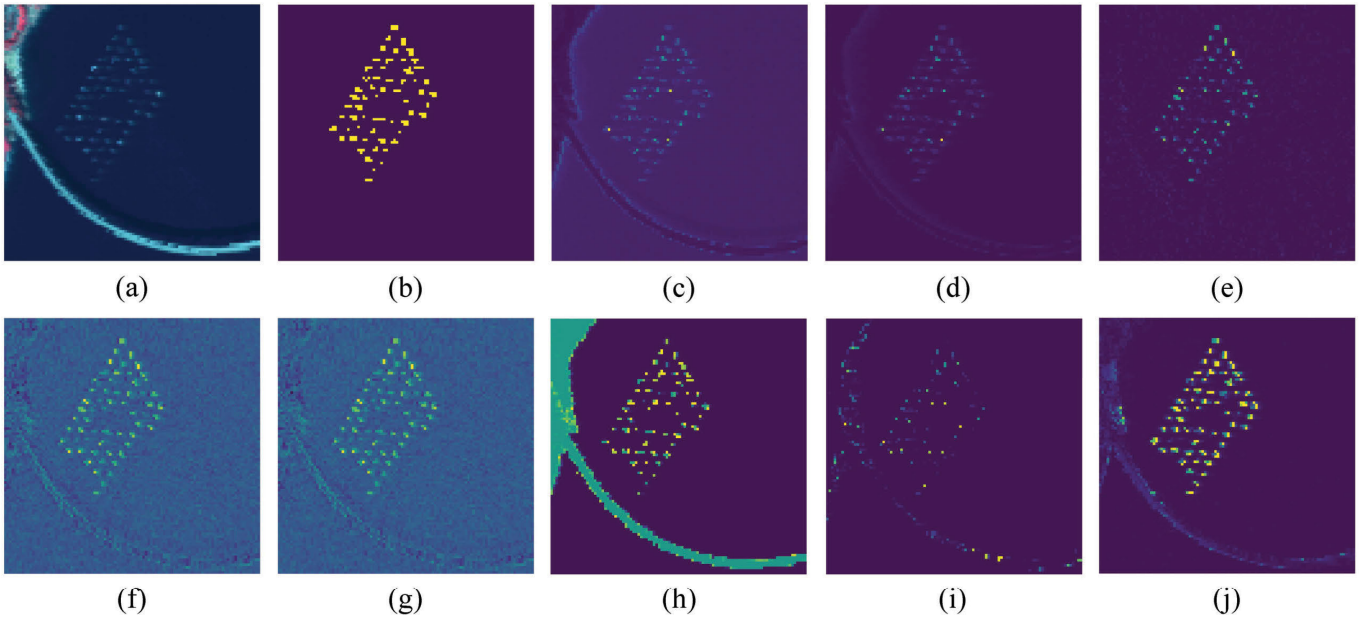
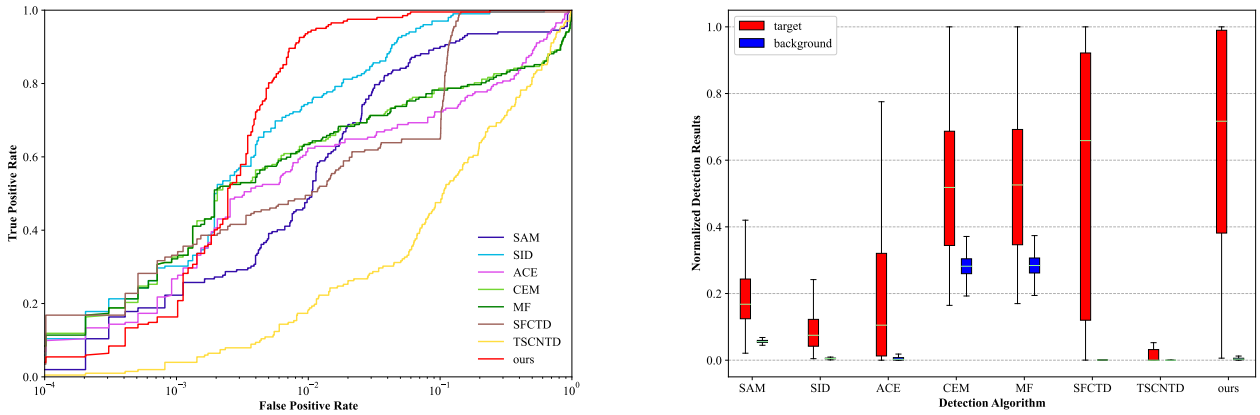


Fig. 10. Beach scene and detection maps of different comparison methods. (a) False color image. (b) Ground truth. (c) SAM. (d) SID. (e) ACE. (f) CEM. (g) MF. (h) SFCTD. (i) TSCNTD. (j) Ours.



(a) (b)

Fig. 11. ROC curves and separability map on beach. (a) ROC curves. (b) Separability map.

from the blue box, which means that targets and backgrounds can be separated to some extent. The ROC curve of our proposed TSTTD shows a competitive TPR at most FPRs, especially around 10^{-2}. Then the AUC values are provided in Table I to quantitatively assess the detection performance. The AUC value of TSTTD only reaches 0.76545. Our proposed TSTTD achieves the best AUC value 0.99507, which outperforms 0.98373 of the second-best SID by 0.01134.

For San Diego, the false color image and ground-truth map are shown in Fig. 12(a) and (b), and the maps of detection results for different detectors are shown in Fig. 12(c)–(j). We find that the highlighting effect of the first five classic detectors on targets is obviously worse than SFCTD, TSCNTD, and TSTTD, and they fail to clearly locate the boundary of targets. Although three classical methods, ACE, CEM, and MF, are able to emphasize several pixels in the middle of the airplane on the right, they ignore pixels in the edge area. SID achieves the worst detection performance as it ignores almost all the target pixels, failing to emphasize on targets. SFCTD achieves good performance in separating the targets from backgrounds, but the shapes of airplanes are not well-defined. TSCNTD can better highlight the airplane on the right, but it fails to describe the shapes of the other two airplanes and mistakenly identify backgrounds near targets. In contrast, our proposed method, TSTTD, not only emphasizes all the targets with extremely high confidence but also clearly describes the shape of targets. The corresponding separability map and ROC curves are presented in Fig. 13. In the separability map, SID and ACE have the flattest blue boxes, but the red boxes partially overlap with the blue boxes, showing limited ability to separate backgrounds and targets. Our proposed TSTTD has the highest red box and the largest gap between red box and blue box, which indicates the excellent performance in target detection and target–background separation. According to the

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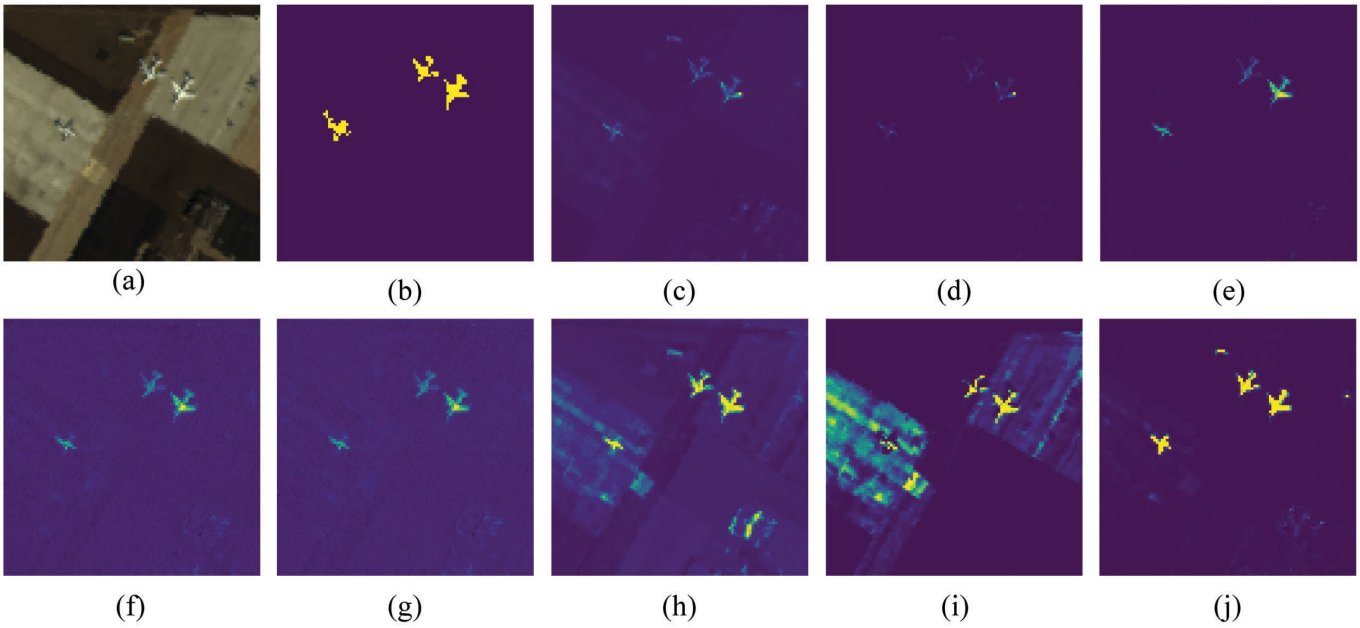


Fig. 12. San Diego scene and detection maps of different comparison methods. (a) False color image. (b) Ground truth. (c) SAM. (d) SID. (e) ACE. (f) CEM. (g) MF. (h) SFCTD. (i) TSCNTD. (j) Ours.

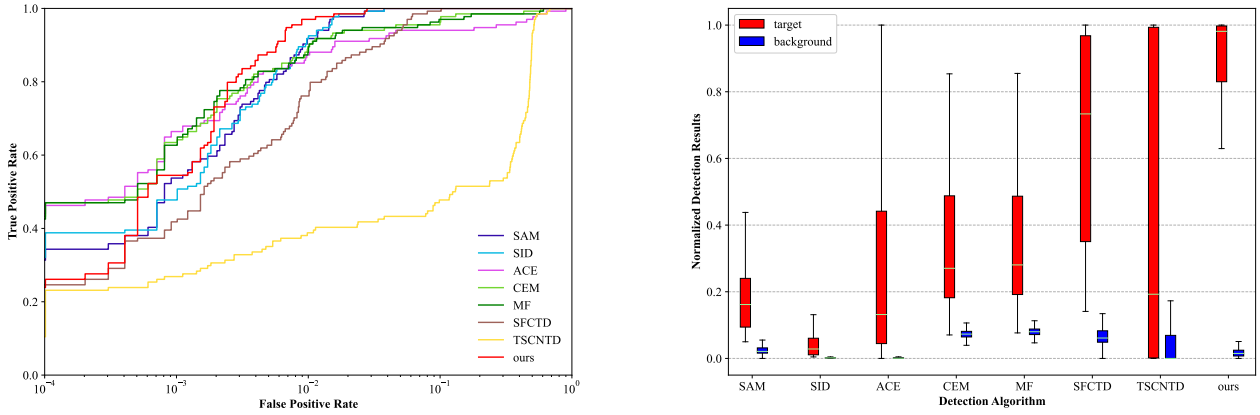


Fig. 13. ROC curves and separability map on San Diego. (a) ROC curves. (b) Separability map.

ROC curve of our proposed TSTTD, the detection rate under the low FPR ($10^{-4} - 0$) is competitive. The AUC values in Table I are quantitatively assessing the detection performance. The classical SID achieves high AUC value 0.99684. The high AUC value of SID owes to the fact that a very large proportion of pixels are suppressed, but the visibility of targets in the detection map of SID is terribly low. Our proposed TSTTD achieves the best AUC value 0.99787 and outperforms the second-best SID of 0.99684.

For HYDICE, the false color image and ground-truth map are shown in Fig. 14(a) and (b), and the maps of detection results for different detectors are shown in Fig. 14(c)–(j). The first five detectors are virtually invalid in locating all the targets except for the upper right. In contrast, SFCTD, TSCNTD, and TSTTD achieve effective performance in locating and highlighting the target pixels. However, TSCNTD mistakenly detects the strong edges of roads as targets, and thus, the overall detection results are seriously and negatively affected by false alarms. SFCTD performs better than TSCNTD in suppressing backgrounds, but there are still pixels at the edge of roads that are incorrectly identified as targets. In contrast, TSTTD shows excellent suppression at backgrounds and exhibits high visual contrast. The corresponding separability map and ROC curves are presented in Fig. 15. In the separability map, the red box of our proposed TSTTD has the highest quartile, which means the excellent performance in highlighting targets. Moreover, the blue and red boxes of TSTTD are completely separate with extremely large gap, indicating separation of targets from backgrounds. The ROC curve of our proposed TSTTD is the closest to the top-left corner, and the detection rate is competitive under the low FPR from 10^{-4} to 0. When the TPR equals 1, that is, all the targets are truly detected, the false alarm rate of TSTTD is the lowest and far below 10^{-2} which indicates

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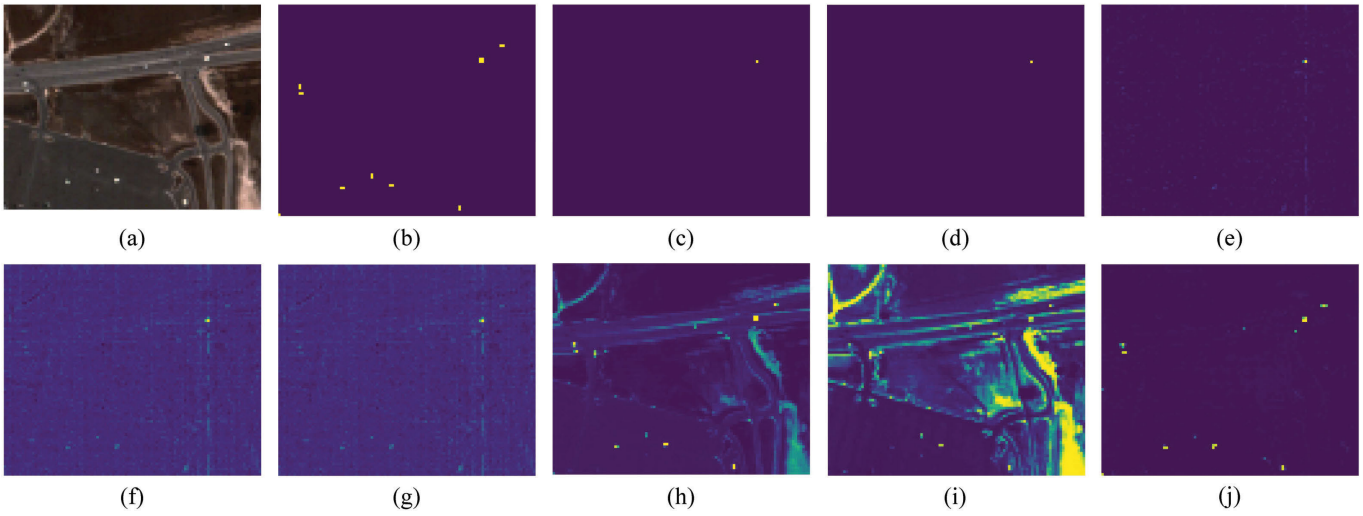


Fig. 14. HYDICE scene and detection maps of different comparison methods. (a) False color image. (b) Ground truth. (c) SAM. (d) SID. (e) ACE. (f) CEM. (g) MF. (h) SFCTD. (i) TSCNTD. (j) Ours.

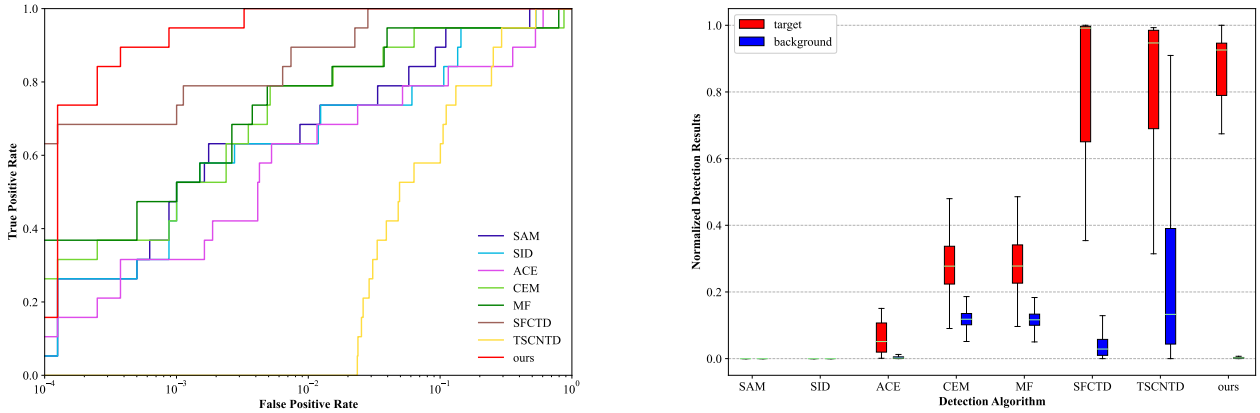


Fig. 15. ROC curves and separability map on HYDICE. (a) ROC curves. (b) Separability map.

the most excellent background suppression. Then the AUC values are provided in Table I to quantitatively assess the detection performance. Our proposed TSTTD achieves the best AUC value 0.99966, outperforming 0.99648 of the second-best learning-based SFCTD by 0.00318. Another learning-based detector, TSCNTD, only reaches the lowest AUC value of 0.88615 due to the limited background suppression.

A summary of the above experiments is as follows. Analyzing the above comparison experiments of eight detectors for multiple datasets, the following conclusions can be concluded. SAM, SID, and ACE work well in suppressing backgrounds in datasets apart from the muffle dataset, but they fail to highlight targets and are unable to describe the shape of targets. The performance of CEM and MF is similar on each dataset; they can locate targets and describe the shapes approximately, but they are strongly influenced by noise that yields poor background suppression. In summary, the first five classical detectors are limited in complex practical application scenarios. The deep-learning-based methods for comparison, SFCTD and TSCNTD, always locate and highlight target pixels well, but they always mistakenly detect backgrounds with high detection scores, resulting in high false alarm rates. Our proposed TSTTD outperforms SFCTD and TSTTD due to the strong ability in spectral feature extraction. The comparison experiments prove the superior and stable detection performance of TSTTD in all the six datasets. In conclusion, TSTTD achieves an outstanding balance between targets' detection and backgrounds' suppression.

E. Model Analysis

To verify the effectiveness of these major components of the developed method, i.e., SCF, GSE, and ISIA loss, we analyze three variants of our proposed TSTTD. For that, ablation experiments are conducted on six hyperspectral images. In addition to the component to be analyzed, the other settings of the variants are consistent with the proposed method.

Fig. 16 presents the detection results of the developed method and its three variants. Table II shows the AUC values of model analysis, which qualitatively measure the performance of the detectors. Fig. 17 shows the ROC curves of model analysis, which qualitatively evaluate the performance of the detectors.

1) Analysis of Skip-Layer Concatenation Fusion: Fig. 16(a) shows the detection maps of the TSTTD variant without SCF.

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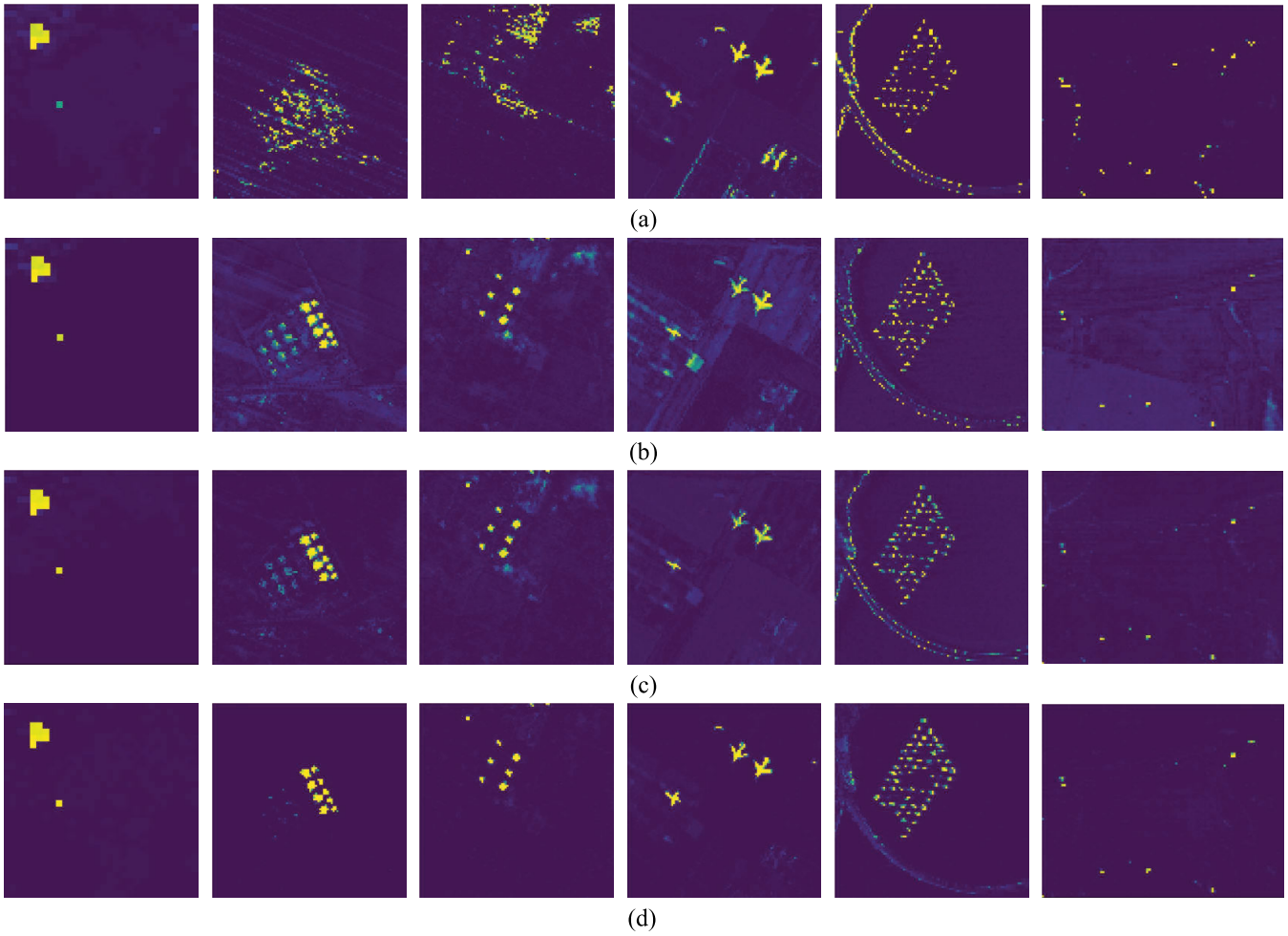


Fig. 16. Detection maps for model analysis of the proposed TSTTD on six HSIs. (a) Without SCF. (b) Without group-based embedding. (c) BCE loss variant. (d) Proposed TSTTD.

TABLE II
AUC VALUES FOR MODEL ANALYSIS

Methods	Muufl Gulfport	Urban1	Urban2	Beach	San Diego	HYDICE
w/o skip-layer concatenation fusion	0.98966	0.88975	0.90265	0.95396	0.99416	0.99380
w/o group-based embedding	0.98079	0.99711	0.99940	0.95032	0.99180	0.99148
BCE loss	0.97124	0.99700	0.99908	0.98338	0.99453	0.99937
proposed TSTTD	**0.99843**	**0.99787**	**0.99958**	**0.99507**	**0.99787**	**0.99966**

The variant without SCF has extremely poor performance on Urban1 and Urban2, which fails to locate and highlight target pixels. In the case of other images, although the variant can locate and highlight targets, many background pixels are incorrectly detected with high scores. By comparison, the proposed TSTTD with SCF outperforms the variant without SCF. Moreover, the ROC curves of our proposed TSTTD are much closer to the upper left than that without skip-layer fusion and the AUC values are higher. Therefore, the proposed TSTTD with SCF can achieve better performance, which demonstrates the necessity and effectiveness of skip-layer fusion.

2) Analysis of Group-Based Spectral Embedding: In this ablation experiment, the single-band embedding (SBE) replaces the GSE. Fig. 16(b) shows detection maps of the TSTTD variant without GSE. As shown, compared with our proposed method using GSE, the variant using SBE is inferior in suppressing backgrounds and has more undesired false alarm pixels. The ROC curves of our proposed TSTTD which use GSE outperform the variant which uses SBE, as the green curves leading the blue curves. Although for some images the blue curves are ahead of the green curves at the beginning, they fall behind afterward. Therefore, our proposed TSTTD can achieve better performance that demonstrates the validity of the GSE. The above conclusion can also be proved by the AUC values in Table II.

3) Analysis of Loss Function: Fig. 16(c) shows detection maps of the TSTTD variant without the designed ISIA loss. Compared with our proposed ISIA loss which joints the HNM triplet loss and the BCE loss, the variant using only BCE

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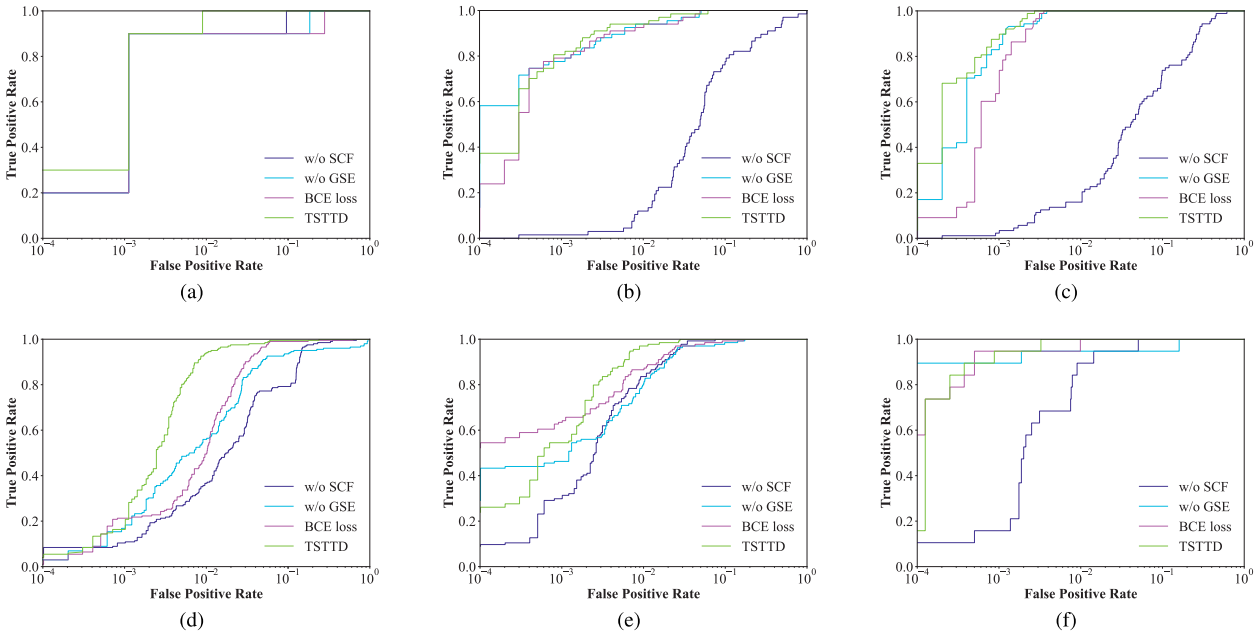


Fig. 17. ROC curves for the model analysis of the proposed TSTTD on six HSIs. (a) MUUFL Gulfport. (b) Urban1. (c) Urban2. (d) Beach. (e) San Diego. (f) HYDICE.

loss is inferior in suppressing some background pixels. For Urban1 and HYDICE, although there is a partial overlap between the ROC curves of TSTTD using ISIA loss and the variant using only BCE loss, the AUC values of the proposed TSTTD with ISIA loss are the highest. For San Diego, the wine curve is ahead of the green curve at the beginning, but it lags behind afterward. For other images, the green curves lead the wine curves. Therefore, the proposed ISIA loss can achieve better performance on the whole, which can also be proved by the AUC values in Table II. The above results demonstrate the validity of the ISIA loss, which helps improve the separability between the targets and backgrounds thus improving the detection performance. In addition to ISIA loss and BCE loss, support vector machines (SVMs) with kernel trick are also available for network parameter optimization. SVM with the kernel function can provide a decision hyperplane parameterized by the Lagrange multiplier, separating positives from negatives. Then an SVM-based loss can be formulated using the Lagrange multiplier and the kernel function. When minimizing the SVM-based loss, it seeks a representation map that improves the similarity between positive pairs and the dissimilarity between negative pairs, achieving a similar effect as our proposed ISIA loss.

Analyzing the ablation experiments above, the skip-layer fusion, group-based embedding, and ISIA loss are shown to be effective and necessary in highlighting targets and suppressing backgrounds, which can improve the performance of our proposed network for target detection.

IV. Conclusion

In this article, we propose a triplet spectralwise transformer target detector (TSTTD) for HTD, which integrates the triplet architecture and transformer. First, the triplet spectralwise transformer network is designed with the GSE and SCF. These two modules help learn local features from multiple adjacent bands and global features with long-range dependence, and the ablation experiments show their necessity and effectiveness. Then, considering the limitation of training data, this work explores a data augmentation technique to generate sufficient and balanced training samples for balanced learning. Moreover, a novel ISIA loss is proposed for improving separability between the targets and backgrounds. The ISIA loss helps distinguish between positive and negative sample pairs and improves the detection performance better than the BCE loss, as demonstrated by ablation experimental results. The effectiveness of our proposed TSTTD is demonstrated by greatly suppressing background pixels and highlighting target pixels, as well as obtaining the comparable or even better performance when compared with the state-of-the-art methods on all the six datasets.

Here are some issues that merit further exploration in the future. First, it is worth exploring more robust algorithms for handling high dimensionality. Then, incorporating spatial information and spectral information would also be an interesting future direction. Furthermore, we intend to incorporate more physical characteristics of HSIs into deep models, thereby improving the interpretability and scientific validity of the designed algorithms. In addition, considering specific applications, we would like to explore the TSTTD's detection capability for the same targets appearing in different HSIs under different environmental conditions.

References

[1] D. Hong et al., "Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021.

[2] P. Zhong, Z. Gong, and J. Shan, "Multiple instance learning for multiple diverse hyperspectral target characterizations," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 1, pp. 246–258, Jan. 2020.

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[3] Z. Gong, P. Zhong, Y. Yu, W. Hu, and S. Li, "A CNN with multiscale convolution and diversified metric for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3599–3618, Jun. 2019.

[4] Y. Yuan, J. Lin, and Q. Wang, "Hyperspectral image classification via multitask joint sparse representation and stepwise MRF optimization," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 2966–2977, Dec. 2016.

[5] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[6] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[7] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[8] D. Manolakis, D. Marden, and G. A. Shaw, "Hyperspectral image processing for automatic target detection applications," *Lincoln Lab. J.*, vol. 14, no. 1, pp. 79–116, 2003.

[9] J. Theiler, A. Ziemann, S. Matteoli, and M. Diani, "Spectral variability of remotely sensed target materials: Causes, models, and strategies for mitigation and robust exploitation," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 8–30, Jun. 2019.

[10] A. Zare and K. C. Ho, "Endmember variability in hyperspectral analysis: Addressing spectral variability during spectral unmixing," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 95–104, Jan. 2014.

[11] E. J. Ientilucci and S. Adler-Golden, "Atmospheric compensation of hyperspectral data: An overview and review of in-scene and physics-based approaches," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 31–50, Jun. 2019.

[12] Y. Cohen, Y. August, D. G. Blumberg, and S. R. Rotman, "Evaluating subpixel target detection algorithms in hyperspectral imagery," *J. Electr. Comput. Eng.*, vol. 2012, pp. 1–15, Jan. 2012.

[13] J. E. West, D. W. Messinger, E. J. Ientilucci, J. P. Kerekes, and J. R. Schott, "Matched filter stochastic background characterization for hyperspectral target detection," *Proc. SPIE*, vol. 5806, pp. 1–12, Jun. 2005.

[14] R. A. Schowengerdt, *Remote Sensing: Models and Methods for Image Processing*, 3rd ed. Orlando, FL, USA: Elsevier, 2007.

[15] C.-I. Chang, "An information-theoretic approach to spectral variability, similarity, and discrimination for hyperspectral image analysis," *IEEE Trans. Inf. Theory*, vol. 46, no. 5, pp. 1927–1932, Aug. 2000.

[16] D. Manolakis and G. Shaw, "Detection algorithms for hyperspectral imaging applications," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 29–43, Jan. 2002.

[17] D. Manolakis, R. Lockwood, T. Cooley, and J. Jacobson, "Is there a best hyperspectral detection algorithm," *Proc. SPIE*, vol. 7334, Apr. 2009, Art. no. 733402.

[18] E. J. Kelly, "An adaptive detection algorithm," *IEEE Trans. Aerosp. Electron. Syst.*, vol. AES-22, no. 2, pp. 115–127, Mar. 1986.

[19] S. Kraut and L. L. Scharf, "The CFAR adaptive subspace detector is a scale-invariant GLRT," *IEEE Trans. Signal Process.*, vol. 47, no. 9, pp. 2538–2541, Sep. 1999.

[20] S. Kraut, L. L. Scharf, and R. W. Butler, "The adaptive coherence estimator: A uniformly most-powerful-invariant adaptive detection statistic," *IEEE Trans. Signal Process.*, vol. 53, no. 2, pp. 427–438, Feb. 2005.

[21] F. C. Robey, D. R. Fuhrmann, E. J. Kelly, and R. Nitzberg, "A CFAR adaptive matched filter detector," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 28, no. 1, pp. 208–216, Jan. 1992.

[22] W. Farrand, "Mapping the distribution of mine tailings in the Coeur d'Alene River Valley, Idaho, through the use of a constrained energy minimization technique," *Remote Sens. Environ.*, vol. 59, no. 1, pp. 64–76, Jan. 1997.

[23] C.-I. Chang, H. Ren, and M. Hsueh, "Revisiting the target-constrained interference-minimized filter (TCIMF)," *Proc. SPIE*, vol. 5159, pp. 339–348, Jan. 2004.

[24] Z. Gong, P. Zhong, and W. Hu, "Statistical loss and analysis for deep learning in hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 322–333, Jan. 2021.

[25] X. He, Y. Chen, and Z. Lin, "Spatial–spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[26] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[27] Z. Li and Y. Zhang, "Hyperspectral anomaly detection based on improved RX with CNN framework," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 2244–2247.

[28] H. Su, Z. Wu, H. Zhang, and Q. Du, "Hyperspectral anomaly detection: A survey," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 64–90, Mar. 2022.

[29] M. Hu, C. Wu, L. Zhang, and B. Du, "Hyperspectral anomaly change detection based on autoencoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3750–3762, 2021.

[30] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[31] S. Chopra, R. Hadsell, and Y. LeCun, "Learning a similarity metric discriminatively, with application to face verification," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jul. 2005, pp. 539–546.

[32] G. Koch, R. Zemel, and R. Salakhutdinov, "Siamese neural networks for one-shot image recognition," in *Proc. ICML Deep Learn. Workshop*, vol. 2, Jul. 2015, pp. 1–8.

[33] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," 2020, *arXiv:2002.05709*.

[34] S. Zagoruyko and N. Komodakis, "Learning to compare image patches via convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4353–4361.

[35] X. Zhang, K. Gao, J. Wang, Z. Hu, H. Wang, and P. Wang, "Siamese network ensembles for hyperspectral target detection with pseudo data generation," *Remote Sens.*, vol. 14, no. 5, p. 1260, Mar. 2022.

[36] D. Z. Anderson, J. D. Zollweg, and B. J. Smith, "Paired neural networks for hyperspectral target detection," *Proc. SPIE*, vol. 11139, Sep. 2019, Art. no. 111390.

[37] D. Zhu, B. Du, and L. Zhang, "Two-stream convolutional networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6907–6921, Aug. 2021.

[38] W. Rao, L. Gao, Y. Qu, X. Sun, B. Zhang, and J. Chanussot, "Siamese transformer network for hyperspectral image target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526419.

[39] G. Zhang, S. Zhao, W. Li, Q. Du, Q. Ran, and R. Tao, "HTD-Net: A deep convolutional neural network for target detection in hyperspectral imagery," *Remote Sens.*, vol. 12, no. 9, p. 1489, May 2020.

[40] Y. Gao, Y. Feng, and X. Yu, "Hyperspectral target detection with an auxiliary generative adversarial network," *Remote Sens.*, vol. 13, no. 21, p. 4454, Nov. 2021.

[41] S. Matteoli, E. J. Ientilucci, and J. P. Kerekes, "Operational and performance considerations of radiative-transfer modeling in hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 4, pp. 1343–1355, Apr. 2011.

[42] J. Kim, J. Kim, and J. Joung, "Siamese hyperspectral target detection using synthetic training data," *Electron. Lett.*, vol. 56, no. 21, pp. 1116–1118, Oct. 2020.

[43] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[44] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[45] M. Raghu et al., "Do vision transformers see like convolutional neural networks?" in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 1–13.

[46] B. Liu, R. Tang, Y. Chen, J. Yu, H. Guo, and Y. Zhang, "Feature generation by convolutional neural network for click-through rate prediction," in *Proc. World Wide Web Conf.*, May 2019, pp. 1119–1129.

[47] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, I. Guyon, Ed. Red Hook, NY, USA: Curran Associates, 2017, pp. 5998–6008.

[48] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[49] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

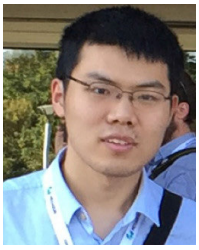
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 05:47:56 UTC from IEEE Xplore. Restrictions apply.

[50] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 815–823.
[51] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollar, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[52] H. O. Song, Y. Xiang, S. Jegelka, and S. Savarese, "Deep metric learning via lifted structured feature embedding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4004–4012.
[53] Y. Wen, K. Zhang, Z. Li, and Y. Qiao, "A discriminative feature learning approach for deep face recognition," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2016, pp. 499–515.
[54] E. Simo-Serra, E. Trulls, L. Ferraz, I. Kokkinos, P. Fua, and F. Moreno-Noguer, "Discriminative learning of deep convolutional feature point descriptors," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 118–126.
[55] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[56] J. Wang et al., "Learning fine-grained image similarity with deep ranking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2014, pp. 1386–1393.
[57] V. Balntas, E. Riba, D. Ponsa, and K. Mikolajczyk, "Learning local feature descriptors with triplets and shallow convolutional neural networks," in *Proc. Brit. Mach. Vis. Conf.*, 2016, pp. 1–12.
[58] G. Paul, Z. Alina, C. Ryan, A. Jen, and T. Grady, "MUUFL Gulfport hyperspectral and LiDAR airborne data set," Univ. Florida, Gainesvill, FL, USA, Tech. Rep., REP-2013-570, 2013.
[59] X. Du and A. Zare, "Technical report: Scene label ground truth map for MUUFL Gulfport data set," Univ. Florida, Gainesville, FL, Tech. Rep., 20170417, Apr. 2017.
[60] X. Song, L. Zou, and L. Wu, "Detection of subpixel targets on hyperspectral remote sensing imagery based on background endmember extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2365–2377, Mar. 2021.
[61] Y. Zhang, K. Wu, B. Du, and X. Hu, "Multitask learning-based reliability analysis for hyperspectral target detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 7, pp. 2135–2147, Jul. 2019.
[62] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2016.
[63] C. Ferri, J. Hernndez-Orallo, and P. Flach, "A coherent interpretation of AUC as a measure of aggregated classification performance," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2011, pp. 657–664.



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